

Perceived Stress Prediction Using Machine Learning with Feature Selection on EMA Stress and Sleep Data: A StudentLife Study

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Abstract—Stress is the most commonly reported mental health concern among college students, but monitoring it in a continuous and unobstructed manner is not possible. In our work, we explore how weekly EMA-based stress reports and smartphone-inferred sleep data are enough to predict a student’s perceived stress level (PSS). We built and compared five classifiers—Logistic Regression, SVM-RBF, Random Forest, XGBoost, and a lightweight tabular transformer—using 12 manually engineered cross-modal features derived from EMA data and sleep statistics. Mutual information ranking with an incremental feature selection strategy was used to guide our feature selection, and classification thresholds were adjusted accordingly to achieve balanced accuracy. Using information from the StudentLife dataset (48 students, 10 weeks), XGBoost was able to achieve a ROC-AUC of 0.734 with just 3 features and a balanced accuracy of 0.694. These results helped us conclude that feature interactions combining stress and sleep signals output more predictive value than when used individually.

Keywords—Perceived Stress Prediction, Machine Learning, Feature Selection, Mutual Information, Incremental Feature Selection, EMA (Ecological Momentary Assessment), Sleep Data, Cross-Modal Features, XGBoost, Transformer Models

I. INTRODUCTION

College students are highly affected by psychological stress, driven by academic pressure, social changes, and life transitions. Clinical screenings are infrequent and fail to capture weekly stress variations. Smartphones, carried continuously, provide an alternative by passively capturing behavioral signals reflecting mental state. The StudentLife dataset contains EMA responses, sleep, activity, and location data from 48 students over 10 weeks. Prior work shows a strong relationship between stress and sleep, but many studies treat them independently, missing cross-modal interactions.

This work evaluates weekly aggregated EMA-based stress and smartphone-derived sleep features to predict stress levels measured by PSS. Daily data is converted into weekly mean and variability features to reduce noise. Cross-modal feature engineering includes interaction and ratio-based features.

Mutual Information ranking with Incremental Feature Selection identifies minimal feature subsets. Multiple models, including Logistic Regression, SVM-RBF, Random Forest, XGBoost, and

a Transformer, are evaluated using stratified cross-validation with threshold optimization [2].

Results show cross-modal features improve performance, with XGBoost achieving the best results..

II. DATASET AND PREPROCESSING

The StudentLife study conducted during Spring 2013 at Dartmouth College included 48 participants (30 undergraduate, 18 graduate; 38 male, 10 female) enrolled in a smartphone programming course. All participants carried Android phones for 10 weeks, which collected accelerometer, audio, GPS, and sleep data, and prompted EMA responses 5–6 times per day [1]. EMA stress scores (1–5) were aggregated into weekly mean and standard deviation (stress_mean, stress_std). Sleep data, inferred from phone sensors, provided weekly mean and standard deviation of sleep hours and quality (4 features).

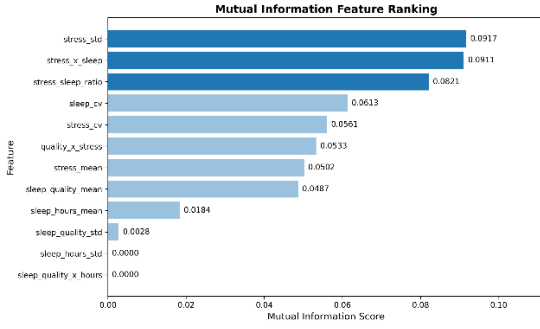
Ground truth labels were derived from post-study PSS (0–40), split at the median into high-stress (1) and low-stress (0), ensuring near-balanced classes [2]. EMA and sleep data were merged using student ID and week. Rows with missing or non-numeric values were removed, and standard deviation was set to zero for weeks with a single observation.

III. FEATURE ENGINEERING AND SELECTION

A filter-based feature selection approach used Mutual Information (MI) to rank 12 engineered features based on relevance to binary stress labels, with missing values handled using median imputation. MI measures dependency between feature X and target Y ($I(X; Y) = H(Y) - H(Y | X)$), where higher scores indicate greater relevance and values near zero indicate low contribution [3]. Feature engineering combined statistical, variability-based, and cross-modal features. Weekly statistics included stress_mean, stress_std, sleep_hours_mean, sleep_hours_std, sleep_quality_mean, and sleep_quality_std. Variability was captured using stress_cv and sleep_cv. Cross-modal features such as stress_sleep_ratio, stress_x_sleep, quality_x_stress, and sleep_quality_x_hours modeled interactions between stress and sleep.

Results show stress_std, stress_x_sleep, and stress_sleep_ratio as most informative, while sleep_hours_std and sleep_quality_x_hours contribute minimally. Cross-modal features consistently ranked higher. Incremental Feature

Selection (IFS) constructed nested subsets (Top-1 to Top-12) to identify the smallest feature set achieving optimal performance, improving efficiency and interpretability [4].



IV. METHODOLOGY

A. Classical Models

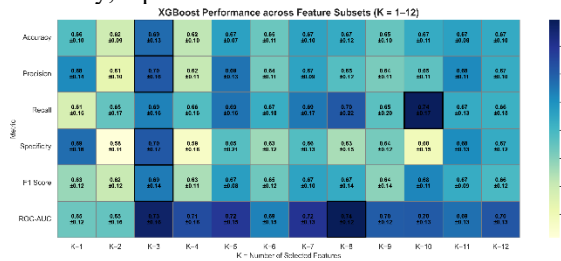
We evaluated Logistic Regression, SVM (RBF), Random Forest, and XGBoost using a standardized pipeline with median imputation, feature scaling, and randomized hyperparameter tuning with stratified cross-validation. Class weights addressed imbalance. Performance was assessed using 10-fold stratified cross-validation with threshold tuning to optimize F1-score and balanced accuracy.

B. Transformer Model

The principal contribution is a custom Transformer model for tabular mental health prediction, where each feature is treated as a token and embedded into a higher-dimensional space to capture relationships between stress and sleep through self-attention. The architecture includes feature-wise embedding, positional encoding, multi-head attention, encoder blocks, and a classification head. Implemented in PyTorch, it uses AdamW, binary cross-entropy, class balancing, early stopping, and gradient clipping, with stratified 5-fold evaluation. The model captures cross-modal interactions and achieves strong recall, though XGBoost performs slightly better due to limited data.

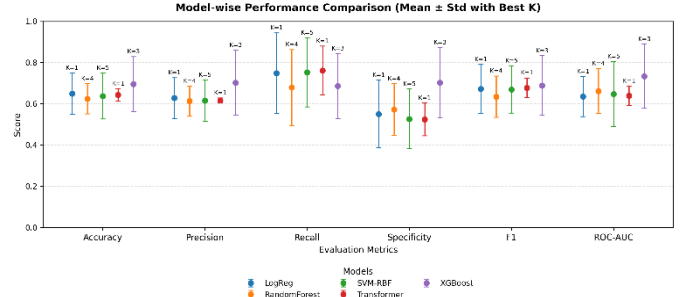
V. RESULTS

An Incremental Feature Selection (IFS) strategy was used to evaluate feature contributions based on Mutual Information ranking [3][4]. Features were added sequentially from most to least informative (Top-1 to Top-12), and both classical models and the proposed Transformer were trained on each subset. Performance was assessed using stratified cross-validation with threshold optimization. Metrics included Accuracy, Precision, Recall, Specificity, F1-score, ROC-AUC, and Balanced Accuracy, reported with mean and standard deviation.



Results show performance varies with feature count, highlighting the need for optimal selection. XGBoost achieved the best overall performance (ROC-AUC ~ 0.74 , F1 ~ 0.69) and remained stable with few features ($K \approx 3 - 5$). Random Forest produced balanced results, while Logistic Regression showed consistent moderate performance.

SVM-RBF achieved the highest recall (~ 0.93) but lower specificity. The Transformer showed competitive recall (~ 0.75) and strong precision, capturing feature interactions, though slightly underperforming XGBoost due to limited dataset size.



VI. CONCLUSION

This study shows that combining EMA-based stress and smartphone-derived sleep features improves mental health prediction, with cross-modal feature engineering enhancing performance. Mutual Information and Incremental Feature Selection reveal that a small feature set ($K \approx 3 - 5$) is sufficient, improving efficiency and interpretability [3]. XGBoost achieves the best overall results for tabular data, while the Transformer captures complex feature interactions with strong recall despite limited data. Future work will focus on larger datasets, temporal modeling, additional sensing modalities, and improved interpretability for real-world applications.

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